

Object Oriented Programming
(CS1004)

Date: May 24th 2024

Course Instructor(s)

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Final Exam

Total Time (Hrs): 3

Total Marks: 100

Total Questions: 4

No

Section

Student Signature

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Please answer all parts of a question together on the answer sheet.

CLO1: Employ dynamic memory allocation and de-allocation.

Question No. 1 (CLO 1, PLO 1, C3)

Marks: 20

- a) Show the output of following piece of codes. Please **show** working in the form of pointer diagram for full credit.

i. <pre>void swap(int*& a, int*& b) { int*temp= a; a=b; b=temp; } void main() { int a=5, b=10; int* ptr1=&a, *ptr2=&b; int ** ptr3= &ptr1; cout<<"Data = "<<**ptr3<<endl; swap(ptr1, ptr2); cout<<"*ptr1 is "<<*ptr1<<endl; cout<<"*ptr2 is "<<*ptr2<<endl; }</pre> ✓	ii. <pre>void SomeFunction(int* arr, int size) { int *p= arr; int *q= arr+size-1; while (p < q) { *p= *q; p=p+2; q--; } } void main() { int nums[]={1, 2, 3, 4, 5, 6, 7, 8, 9,10}; int *ptr= nums; SomeFunction(ptr, 10); for(int i=0; i<10; ++i) cout<<nums[i]<<" "; }</pre> ✓
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b) Consider the following code:

```
int main()
{
    int * arr=0;
    int size;

    cin>>size;

    // function1: dynamically initializes an int array of size given
    allocateMemory(arr, size);

    for(int i=0; i<size;i++)
        cin>>arr[i];

    //function2: explanation given below
    RemoveNumbers(arr, size);

    for(int i=0; i<size;i++)
        cout<<arr[i];

    // function3: deallocates memory
    deallocateMemory(arr);
    return 0;
}
```

You are required to **construct** all 3 functions called in the above main.

The RemoveNumbers function removes all the *negative* integers from the array and reduce its size accordingly. The size of original array is passed as a parameter but should be updated by the function. e.g., if arr is the array shown below and value of size is 8

0	1	2	3	4	5	6	7
-2	5	-2	5	6	-6	11	-92

the function call RemoveNumbers(arr, size) transforms the above array into following and updates its size to 4.

0	1	2	3
5	5	6	11

Note that we are printing array in main after this function call and there should be no errors.

CLO2: Design a program using object-oriented programming paradigm.

Question No. 2 (CLO 2, PLO 1, C5)

Marks: 40

This problem tests your knowledge of classes, composition, inheritance and polymorphism Read the question carefully and **design** the required system accordingly.

- a) [Marks: 10] Declare and implement a class called TableOfContents. It contains a dynamic array of type Content, defined by the following struct:

```
struct Content{  
    string name; //chapter name  
    int pno; //page number of the chapter  
};
```

TableOfContents also contains an integer n for the number of contents. The class also has a constructor TableOfContents(const string & fname) which receives a file name and reads the contents from that file. It is a text file in which the first line contains the number of chapters and then alternate lines contain chapter name and corresponding page number. Following is an example file with only two contents.

```
2  
Introduction to Computers  
1  
Introduction to C++ Programming  
25
```

- b) [Marks : 20] A Book contains a table of contents, an ISBN (which is unique for each book), a price and a title. A book is classified into two main categories i.e., a text book and a general book. A textbook has a discount rate and its price is calculated after taking this into account and a general book is sold on full price.

The system only allows objects of the textbook and general book to be instantiated.

- Provide the number and name of the classes that are required to implement the above system.
 - Provide the declarations for each required class (Only declare the needed methods)
 - Provide Implementation for each declared class. For each class you must implement appropriate constructors and methods only.
- c) [Marks: 10] Give the declaration of a class called BookShop which contains a dynamic array of books and the numbers of books for sale in it. For this class implement the following member method only. (Note: You might assume that all the data is already stored in the array; you don't need to take any input):

printBookDetails(): this function goes through the array of books and prints book title and price

CLO3: Employ operator overloading to redefine the way operator works for user-defined classes.

Question No. 3 (CLO 3, PLO 5, C3)

Marks: 20

You are required to **demonstrate** the working of a class MyList that manages a dynamically growing array of cstrings. Partial class definition, driver program (main) and required output is given below. Implement the class MyList such that the given main program runs successfully. Make sure that your program doesn't consume extra space and there shouldn't be any memory leakage or exceptions in your code.

Note: You can use built-in string functionality like strlen, strcpy etc. where required. You do not need to re-write these functions.

Partial declaration of the class and main program:

```
class MyList{
    //You cannot add or modify DATA members
    char** arr;
    int size;
};

void main()
{
    MyList list1; //Initially list will be empty.
    cout<<"list1= "<<list1; //Displays all the strings in the list
    "Apple" + list1; //Adds new string "Apple" in list1
    cout<<"list1= "<<list1; //Displays all the strings in the list
    "Banana" + list1; //Adds new string "Apple" in list1
    cout<<"list1= "<<list1; //Displays all the strings in the list
    MyList list2; //Initially list will be empty.
    cout<<"list2= "<<list2; //Displays all the strings in the list
    "I love Pakistan" + list2; //Adds new string "I love Pakistan" in list2
    cout<<"list2= "<<list2; //Displays all the strings in the list
    MyList list3;
    list3= list1+list2; //Creates another object of MyList having all the strings
    of list 1 and 2
    cout<<"list3= "<<list3; //Displays all the strings in the list
}
```

Required Output:

```
list1= []
list1=[Apple]
list1=[Apple, Banana]
list2= []
list2=[I love Pakistan]
List3=[Apple, Banana, I love Pakistan]
```

CLO4: Employ templates in C++ to parameterize types in functions and classes.

Question No. 4 (CLO 4, PLO 5, C3)

Marks: 20

- i. **Construct** a template function called isPalindrome that checks if a given array is a palindrome (reads the same backward as forward). Also write a main program to **illustrate** the working of the above function using integer and char data types. ✓
- ii. **Construct** a template class called Container that can store elements of any data type. Include member functions to perform the following operations:
 - a. Insert an element into the container. ✓
 - b. Remove an element from the container. ✓
 - c. Check if a specific element exists in the container. ✓
 - d. Write a program to **illustrate** the working of the Container class using integer, double, and string data types. ✓